

CHAPTER 2

LITERATURE SURVEY

2.1 Photo-realistic Virtual Try-On by Adaptively Generating-Preserving Image Content

Authors: H. Yang, R. Zhang, X. Guo, W. Liu, W. Zuo, P. Luo

Published: IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020

This paper presents ACGPN, a novel framework for generating photo-realistic virtual try-on images, addressing challenges such as large occlusions and complex human poses. ACGPN employs a three-stage approach:

1. **Semantic Layout Generation:** It first predicts the parts of the reference image that will change after the clothing is applied (e.g., arms in a shirt or jacket), guiding the process of the try-on.
2. **Clothing Warping:** The method then warps the target clothing based on the predicted semantic layout, incorporating a second-order difference constraint to maintain consistency and prevent warping artifacts during training.
3. **Inpainting for Content Fusion:** An inpainting module merges the warped clothes with the reference image, adaptively generating realistic and detailed clothing overlays for each segment of the body.

Compared to existing techniques, ACGPN produces more perceptually accurate try-on images, capturing richer details and preserving the texture and visual patterns of the clothes. This makes it particularly effective for generating realistic clothing overlays, even with challenging poses and occlusions.

2.2 VITON: An Image-Based Virtual Try-On Network

Authors: X. Han, Z. Wu, Z. Wu, R. Yu, L.S. Davis

Published: IEEE Conference on Computer Vision and Pattern Recognition, 2018

In this paper, they present the design and implementation of a prototype system of Smart Parking Services based on Wireless Sensor Networks (WSNs) that allows vehicle drivers to effectively find the free parking places. The proposed scheme consists of wireless sensor networks, embedded web-server, central web-server and mobile phone application. In the system, low-cost wireless sensors networks modules are deployed into each parking slot

equipped with one sensor node. The state of the parking slot is detected by sensor node and is reported periodically to embedded web-server via the deployed wireless sensor networks. This information is sent to central web-server using Wi-Fi networks in real-time, and also the vehicle driver can find vacant parking lots using standard mobile devices.

2.3 CP-VTON+: Clothing Shape and Texture Preserving Image-Based Virtual Try-On

Authors: M.R. Minar, T.T. Tuan, H. Ahn, P. Rosin, Y.K. Lai

Published: CVPR Workshops, 2020

CP-VTON+ addresses several key challenges in image-based virtual try-on (VTON) tasks, such as handling diverse human poses and clothing styles. Previous methods often struggled with clothing warping and blending due to issues like distorted warps, misaligned clothing, and incorrect human representations. To tackle these problems, CP-VTON+ introduces innovative techniques to preserve clothing shape and texture throughout the try-on process. By improving the accuracy of clothing-human matching and regularizing transform parameters, CP-VTON+ produces more accurate and natural clothing warps. Additionally, it enhances the blending network, ensuring the clothing remains intact during composition. The method significantly outperforms state-of-the-art approaches, both quantitatively and qualitatively, providing high-quality, photo-realistic results for a wide range of human poses and clothing styles.

2.4 Parser-Free Virtual Try-on via Distilling Appearance Flows

Authors: Yuying Ge, Yibing Song, Ruimao Zhang, Chongjian Ge, Wei Liu, Ping Luo

Published: CVPR, 2021

Virtual try-on systems aim to seamlessly fit a target garment image onto a person's image. Traditional methods rely heavily on human parsing, where segmentation inaccuracies can produce unrealistic images with artifacts. Parsing errors can mislead these models, resulting in flawed try-on outputs. Recent innovations have attempted to address this by using knowledge distillation, where a student network learns from the outputs of a parser-based model. However, this approach limits the quality of the student network's results to the parser's imperfections. To overcome these challenges, a "teacher-tutor-student" knowledge distillation method has been proposed. Unlike previous models, this approach leverages "tutor knowledge" from the parser-based outputs and corrects artifacts using "teacher knowledge" derived from real person images

through self-supervision. Additionally, knowledge distillation is framed around distilling appearance flows between the person and garment images. This method helps establish precise dense correspondences, improving the quality and realism of the virtual try-on outputs, surpassing prior segmentation-dependent techniques.

2.5 RMGN: A Regional Mask Guided Network for Parser-free Virtual Try-on

Authors: Chao Lin, Zhao Li, Sheng Zhou, Shichang Hu, Jialun Zhang, Linhao Luo, Jiarun Zhang, Longtao Huang, Yuan He

Published: 24 April 2022

Virtual try-on(VTON) aims at fitting target clothes to reference person images, which is widely adopted in e-commerce. Existing VTON approaches can be narrowly categorized into Parser-Based(PB) and Parser-Free(PF) by whether relying on the parser information to mask the persons' clothes and synthesize try-on images. Although abandoning parser information has improved the applicability of PF methods, the ability of detail synthesizing has also been sacrificed. As a result, the distraction from original cloth may persisting synthesized images, especially in complicated postures and high resolution applications. To address the aforementioned issue, we propose a novel PF method named Regional Mask Guided Network(RMGN). More specifically, a regional mask is proposed to explicitly fuse the features of target clothes and reference persons so that the persisted distraction can be eliminated. A posture awareness loss and a multi-level feature extractor are further proposed to handle the complicated postures and synthesize high resolution images. Extensive experiments demonstrate that our proposed RMGN outperforms both state-of-the-art PB and PF methods. Ablation studies further verify the effectiveness of modules in RMGN.

2.6 Toward Characteristic-Preserving Image-based Virtual Try-On Network

Authors: Bochao Wang, Huabin Zheng, Xiaodan Liang, Yimin Chen, Liang Lin, Meng Yang

Published: ECCV 2018

Image-based virtual try-on systems aim to overlay in-shop clothing onto a person's image while

maintaining the realism of both the clothing and the person. A major challenge is preserving the key characteristics of the clothing, such as texture, logo, and embroidery, while addressing large spatial misalignments between the input image and target clothing. Prior works, including shape context matching techniques, often compromised fine clothing details due to coarse-to-fine strategies.

Key Contributions:

- **Thin-Plate Spline Transformation:** CP-VTON employs a fully learnable **Geometric Matching Module (GMM)** to transform the target clothing into the most fitting shape. Unlike prior methods relying on point correspondences, this approach ensures seamless alignment while preserving garment identity.
- **Try-On Module:** To enhance realism and avoid boundary artifacts, a composition mask is learned to smoothly integrate the warped clothing with the person's image.
- **Performance:** Extensive experiments on a fashion dataset show CP-VTON surpasses previous methods in generating photo-realistic try-on images with preserved clothing details and better alignment.

2.7 Deep Fashion3D: A Dataset and Benchmark for 3D Garment Reconstruction from Single Images

Authors: Heming Zhu, Yu Cao, Hang Jin, Weikai Chen, Dong Du, Zhangye Wang, Shuguang Cui, Xiaoguang Han

Published: ECCV 2020

This paper addresses the challenges of **high-fidelity 3D garment reconstruction** from single images, a key task in applications like virtual try-on and human digitization. While significant progress has made in modelling unclothed human shapes using statistical models like **SMPL**, the reconstruction of clothed humans and garments remains challenging due to the lack of large-scale datasets.

Key Contributions:

- **Introduction of Deep Fashion3D Dataset:**
 - Largest dataset for 3D garment models to date, with 2078 models across 10 categories and 563 garment instances.
 - Includes rich annotations such as 3D feature lines, body poses, and multi-view real

images, enhancing realism and usability for garment modeling.

- Features random garment posing to simulate diverse clothing deformations.
- **Baseline Approach for Single-View Reconstruction:** Combines mesh and implicit representations to capture intricate garment details. Introduces a novel adaptable template that enables learning for various clothing types within a single network.
- **Benchmark Establishment:** Proposes the dataset as a benchmark for evaluating garment reconstruction systems. Conducts extensive experiments to demonstrate the dataset's value and validate the baseline's performance.

2.8 Fill in Fabrics: Body-Aware Self-Supervised Inpainting for Image-Based Virtual Try-On

Authors: H. Zunair, Y. Gobeil, S. Mercier, A. Ben Hamza

Published: 3rd October 2022

This paper introduces the Fill in Fabrics (FIFA) model, a novel approach to virtual try-on that overcomes limitations in handling complex human poses, body shapes, and clothing structures. The proposed self-supervised framework utilizes a conditional GAN architecture, comprising a Fabricator and a virtual try-on pipeline with a Segmenter, Warper, and Fuser. The Fabricator reconstructs masked clothing images, learning the structural details by "filling in fabrics." These representations are transferred to the Warper, enabling precise alignment and refinement of target clothing. The method also

employs a multi-scale structural constraint to improve fit and maintain clothing details like texture and embroidery. Experiments on the VITON dataset show that FIFA achieves state-of-the-art performance, effectively handling complex poses while producing photorealistic results.

2.9 OOTDiffusion: Outfitting Fusion-Based Latent Diffusion for Controllable Virtual Try-On

Authors: Yuhao Xu, Tao Gu, Weifeng Chen, Chengcai Chen

Published: 4th March 2024

This paper introduces OOTDiffusion, a cutting-edge network architecture for realistic and controllable virtual try-on (VTON). Built upon pretrained latent diffusion models,

OOTDiffusion employs an outfitting UNet to accurately learn garment detail features. Unlike traditional approaches, it eliminates the need for a redundant warping process by aligning garment features directly with the target body using outfitting fusion in the self-attention layers of the denoising UNet. To enhance controllability, the method introduces outfitting dropout, enabling the adjustment of garment feature strength via classifier-free guidance. Experiments on the VITON-HD and Dress Code datasets reveal that OOTDiffusion delivers superior quality, realism, and control compared to existing VTON methods, marking a significant advancement in the field. The source code is publicly available at GitHub.